

Chapter 1

The limit of a Function

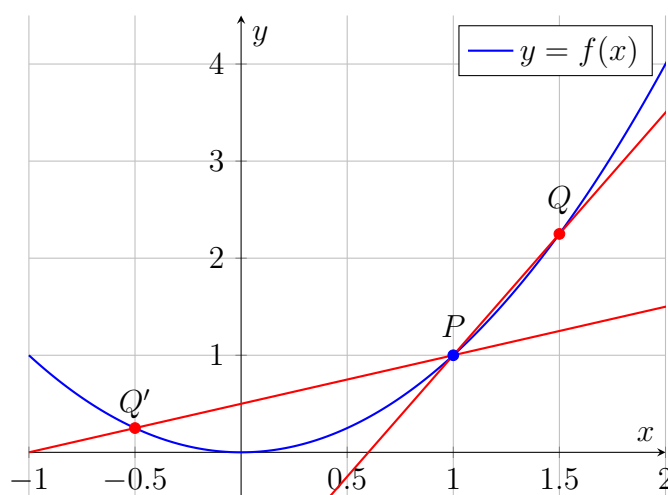
§ Section: 2.1-2.2

1.1 The Tangent and Velocity Problems

In this section, we see how "limits" arise when we aim to find the tangent line to a curve.

Given a curve $y = f(x)$ and a point $P = (x_0, f(x_0))$ on the curve, we want to find the tangent line to the curve $y = f(x)$ at P . First, we can choose a nearby point $Q = (x, f(x))$ and compute the slope of the secant line $\overline{PQ}$ which is $\frac{f(x) - f(x_0)}{x - x_0}$. Then we let the point Q approach P and find the *limit* of the slope $\frac{f(x) - f(x_0)}{x - x_0}$.

We define the slope of the tangent line at P as *the limit of the slopes of the secant lines $\overline{PQ}$* .



Example 1.1

Find the tangent line to the curve $y = x^3$ at the point $(1, 1)$.

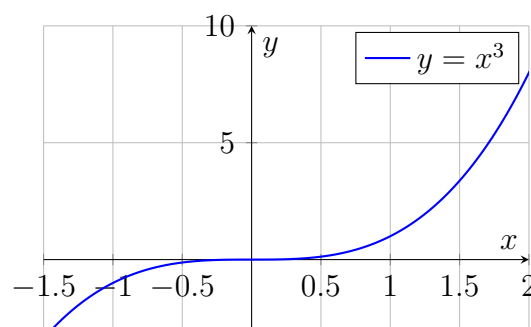
sol: let $Q = (x, x^3)$, $x \neq 1$, and $P = (1, 1)$.

$$\text{Then } m_{\overline{PQ}} = \frac{x^3 - 1}{x - 1} = \frac{(x-1)(x^2+x+1)}{x-1} = x^2 + x + 1.$$

let Q approach P .

Then $m_{\overline{PQ}} \rightarrow 3$ as $x \rightarrow 1$. We say that the tangent line of $y = x^3$ at $(1, 1)$ has slope 3.

Hence the tangent line is $y - 1 = 3(x - 1)$
 $\Rightarrow y = 3x - 2.$

**Exercise 1.1**

Find the tangent line of $y = \sqrt{x}$ at $(1, 1)$.

Marginal Quantities.

In economics, the term *marginal* is used to denote the rate of change of a quantity with respect to its dependent variable. For example,

- **Marginal Cost:** Suppose that $C(x)$ is the total cost function for a manufacturer to produce x units of a commodity. Then the *marginal cost of production* at $x = x_0$ is the limit of

$$\frac{C(x_0 + \Delta x) - C(x_0)}{\Delta x}$$

as Δx approaches 0.

- **Marginal Utility:** Let $U(x)$ be the utility of consuming x units of a commodity. Then, the *marginal utility* is the limit of

$$\frac{U(x_0 + \Delta x) - U(x_0)}{\Delta x}$$

as Δx approaches 0.

Example 1.2

Suppose that the utility of drinking x cups of soda is $U(x) = 3x - \frac{1}{2}x^2$. Find the marginal utility when $x = 2$.

Sol:

$$\begin{aligned} \frac{U(2+\Delta x) - U(2)}{\Delta x} &= \frac{\overbrace{3(2+\Delta x) - \frac{1}{2}(2+\Delta x)^2}^{U(2+\Delta x)} - \overbrace{4}^{U(2)}}{\Delta x} \\ &= \frac{6+3\Delta x - \frac{1}{2}(4+4\Delta x + \Delta x^2) - 4}{\Delta x} \\ &= \frac{\Delta x - \frac{1}{2}\Delta x^2}{\Delta x} = 1 - \frac{1}{2}\Delta x \longrightarrow \downarrow \\ &\hspace{15em} \text{as } \Delta x \rightarrow 0 \end{aligned}$$

1.2 The Limit of a Function

Definition:

We write

$$\lim_{x \rightarrow a} f(x) = L$$

and say

“the limit of $f(x)$, as x approaches a , equals L ”

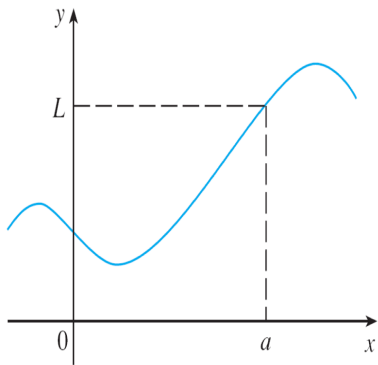
if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a .

Remark:

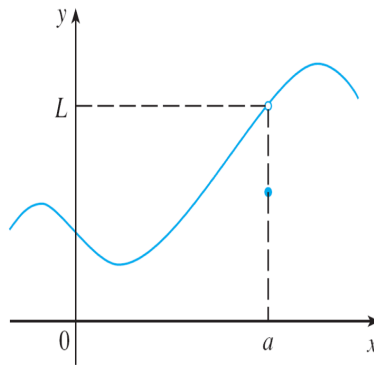
An alternative notation is “ $f(x) \rightarrow L$ as $x \rightarrow a$ ” which is usually read “ $f(x)$ approaches L as x approaches a ”

Remark:

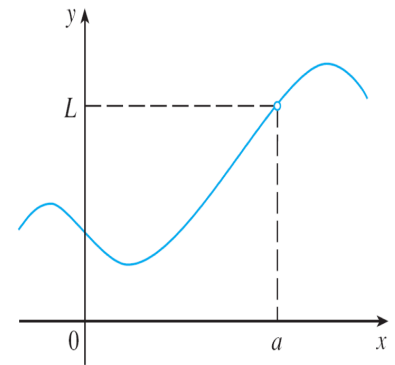
In the definition of limit, we have the phrase $x \neq a$. This means that when finding the limit of $f(x)$ as x approaches a , we never consider how $f(x)$ is defined at $x = a$. In particular, in each case of the following graphs, $\lim_{x \rightarrow a} f(x) = L$.



(a)



(b)



(c)

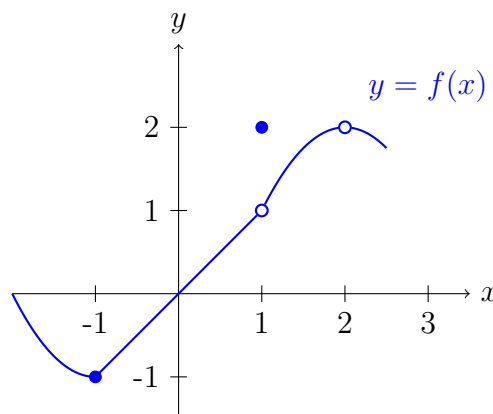
Example 2.1

Determine the values of each limit from the graph below.

$$\lim_{x \rightarrow -1} f(x) = -1 \quad f(-1) = -1$$

$$\lim_{x \rightarrow 1} f(x) = 1 \quad f(1) = 2$$

$$\lim_{x \rightarrow 2} f(x) = 2 \quad f(2) = \text{DNE (Does not exist)}$$

**Corollary:**

If $f(x) = g(x)$ for all $x \neq a$ and $\lim_{x \rightarrow a} f(x) = L$, then the limit of $g(x)$ as x approaches a exists and $\lim_{x \rightarrow a} g(x) = L$.

Example 2.2

Find $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2}$.

sol : $\because \frac{x^2 - 4}{x + 2} = x - 2$ for $x \neq -2$ and $\lim_{x \rightarrow -2} x - 2 = -4$,
 $\therefore \lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2} = -4$.

Example 2.3

Guess the value of $\lim_{x \rightarrow 1} \frac{x - 1}{x^2 - 1} = 0.5$

$x < 1$	$f(x)$
0.5	0.666667
0.9	0.526316
0.99	0.502513
0.999	0.500250
0.9999	0.500025

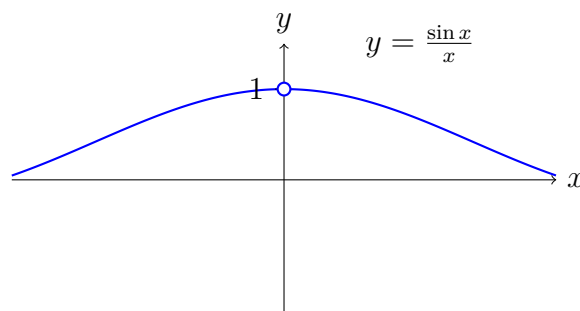
$x > 1$	$f(x)$
1.5	0.400000
1.1	0.476190
1.01	0.497512
1.001	0.499750
1.0001	0.499975

Example 2.4

1. Guess the value of $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 9} - 3}{x^2}$.

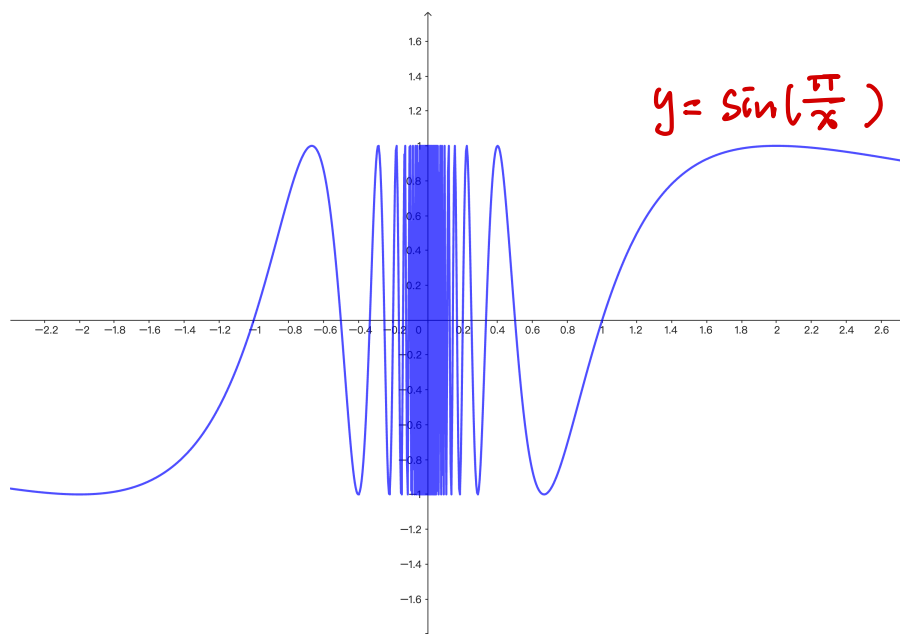
2. Guess the value of $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

3. Investigate $\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right)$. **DNE**





$y = \sin\left(\frac{\pi}{x}\right)$
near
 $x = 0$.

**Example 2.5**

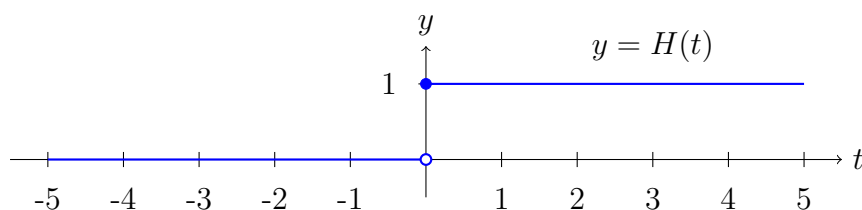
The Heaviside function, $H(t)$, is defined by

$$H(t) = \begin{cases} 0, & \text{if } t < 0 \\ 1, & \text{if } t \geq 0 \end{cases}$$

Investigate $\lim_{t \rightarrow 0} H(t)$.

Sol: $\lim_{t \rightarrow 0} H(t) \text{ DNE}$

$\lim_{t \rightarrow 0^+} H(t) = 1$, $\lim_{t \rightarrow 0^-} H(t) = 0$,



1.3 One-Sided Limits

Definition:

We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say the **left-hand limit** of $f(x)$ as x **approaches** a [or the **limit** of $f(x)$ as x **approaches** a **from the left**] is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a and x less than a .

Remark:

Similarly, if we require that x be greater than a , we get **the right-hand limit** of $f(x)$ as x approaches a is equal to L and we write $\lim_{x \rightarrow a^+} f(x) = L$.

Property:

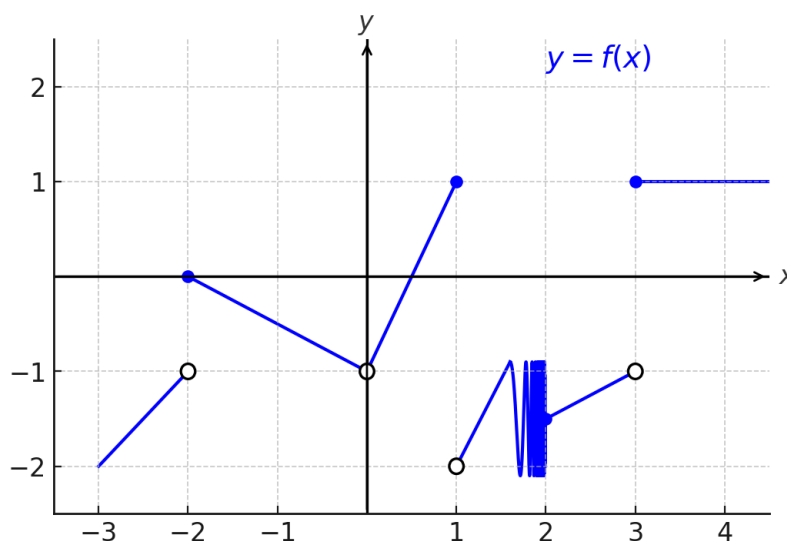
$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L.$$

Example 3.1

Find

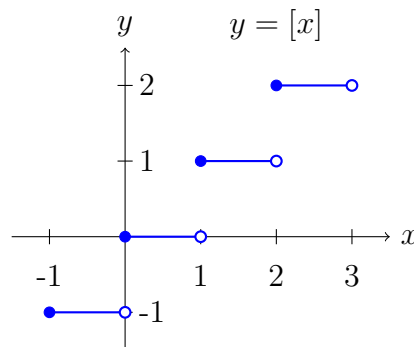
$$\lim_{x \rightarrow -2} f(x) \Leftarrow \begin{cases} \lim_{x \rightarrow -2^-} f(x) = -1 & \lim_{x \rightarrow 1^-} f(x) = 1 & \lim_{x \rightarrow 2^+} f(x) = -1.5 \\ \lim_{x \rightarrow -2^+} f(x) = 0 & \lim_{x \rightarrow 1^+} f(x) = -2 & \lim_{x \rightarrow 3^-} f(x) = -1 \\ \lim_{x \rightarrow 0} f(x) = -1 & \lim_{x \rightarrow 2^-} f(x) = DNE & \lim_{x \rightarrow 3^+} f(x) = 1 \end{cases}$$

DNE



Example 3.2

The floor function (Gauss function): $f(x) = [x]$ = the largest integer that is less than or equal to x .

**Example 3.3**

For all $n \in \mathbb{Z}$, $[n] = n$ $[e] = 2$ $[-\pi] = -4$

Example 3.4

For an $n \in \mathbb{Z}$, find $\lim_{x \rightarrow n^+} [x]$ and $\lim_{x \rightarrow n^-} [x]$.

$\begin{matrix} \text{"} \\ n \end{matrix}$
 $\begin{matrix} \text{"} \\ n-1 \end{matrix}$

Example 3.5

Let $f(x) = [x] + [-x]$. Find $\lim_{x \rightarrow 2^+} f(x)$, $\lim_{x \rightarrow 2^-} f(x)$, $\lim_{x \rightarrow 2} f(x)$, and $f(2)$.

sol: For $x > 2$ and x close to 2, $f(x) = 2 + (-3) = -1$

Hence $\lim_{x \rightarrow 2^+} f(x) = -1$. Similarly, for $x < 2$ and x close to 2, $f(x) = 1 + (-2) = -1$. Hence $\lim_{x \rightarrow 2^-} f(x) = -1$.

Example 3.6

Suppose that $f(x) = \begin{cases} \frac{x^2+x-2}{x+2} & , \text{ if } x > -2, \\ a[\frac{x}{2}] & , \text{ if } x \leq -2. \end{cases}$ Find the constant a such that $\lim_{x \rightarrow -2} f(x)$ exists.

sol: $\lim_{x \rightarrow -2} f(x)$ exists $\Leftrightarrow \lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^+} f(x)$.

$$\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} a \left[\frac{x}{2} \right] = -2a.$$

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} \frac{x^2+x-2}{x+2} = \lim_{x \rightarrow -2^+} (x-1) = -3.$$

$$\left. \begin{array}{l} \Rightarrow -2a = -3 \\ \Rightarrow a = \frac{3}{2} \end{array} \right\}$$

1.4 Infinite Limits

Definition:

Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a , but not equal to a .

Remark:

Another notation for $\lim_{x \rightarrow a} f(x) = \infty$ is “ $f(x) \rightarrow \infty$ as $x \rightarrow a$ ” which is read as “the limit of $f(x)$, as x approaches a , is infinity”

Definition:

Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a , but not equal to a .

Remark:

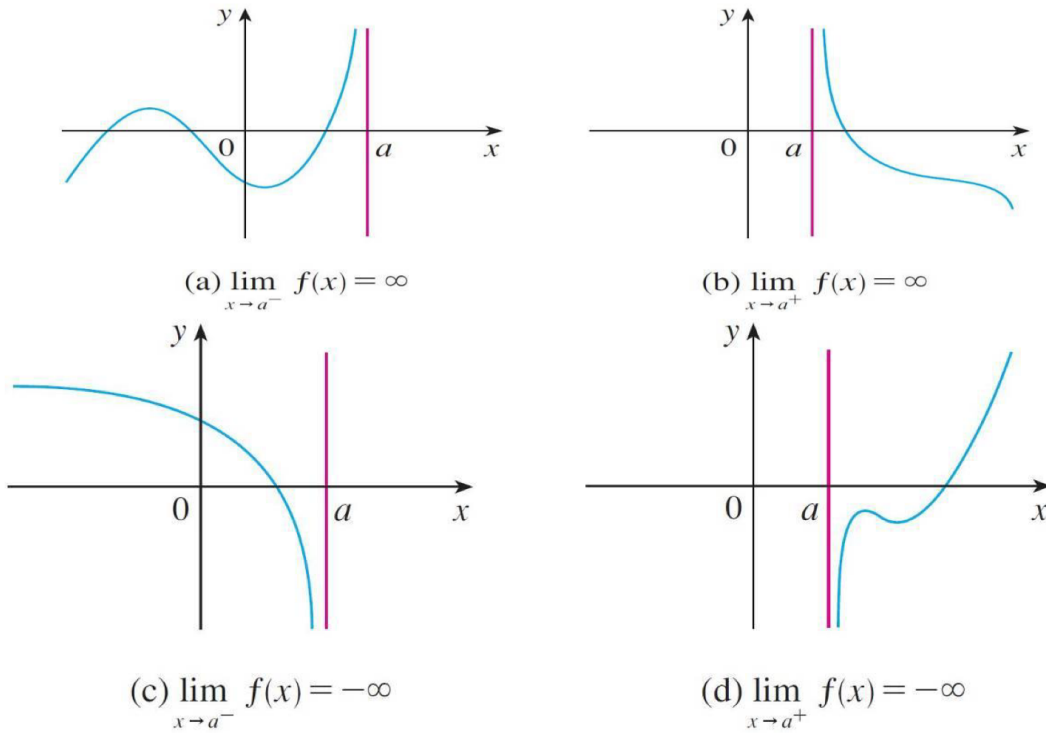
Similar definitions can be given for the one-sided infinite limits

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

**Remark:**

How do we verify that the limit(or one-sided limit) is infinity or negative infinity?

Suppose that $f(x)$ is the quotient of two functions, such as $f(x) = \frac{g(x)}{h(x)}$. Many rational functions and some trigonometric functions (e.g. $\tan x = \frac{\sin x}{\cos x}$, $\sec x = \frac{1}{\cos x}$, etc.) fall into this category. If we can show that $\lim_{x \rightarrow a} g(x) = L \neq 0$ and $\lim_{x \rightarrow a} h(x) = 0$, then the absolute value $|f(x)| = \left| \frac{g(x)}{h(x)} \right|$ tends to ∞ as $x \rightarrow a$. To determine whether the limit is $+\infty$ or $-\infty$, we must carefully analyze the signs of $g(x)$ and $h(x)$ near $x = a$.

Some elementary functions are not written as quotients but they still have infinity limit. For example, $\lim_{x \rightarrow 0^+} \ln x = -\infty$. You should become familiar with the behavior of these functions so that you can recognize and state such limits directly as known facts.

Example 4.1

Find $\lim_{x \rightarrow -1} \frac{1}{(1+x)^2} = \infty$.

Sol: $\because \lim_{x \rightarrow -1} 1 = 1 > 0$, $\lim_{x \rightarrow -1} (1+x)^2 = 0$ and $(1+x)^2 > 0$ for $x \neq -1$.

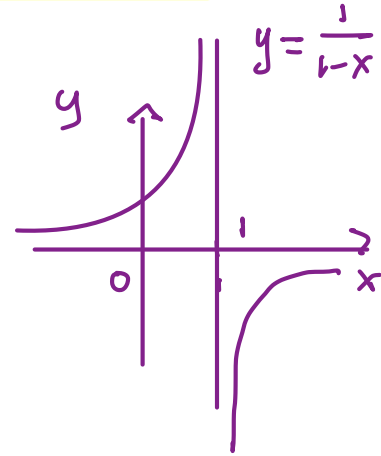
$\therefore \lim_{x \rightarrow -1} \frac{1}{(1+x)^2} = \infty$.

Example 4.2

Find $\lim_{x \rightarrow 1^+} \frac{1}{1-x} = -\infty$. $\lim_{x \rightarrow 1^-} \frac{1}{1-x} = \infty$

Sol: $\because \lim_{x \rightarrow 1^+} 1 = 1 > 0$, $\lim_{x \rightarrow 1^+} (1-x) = 0$. Moreover $1-x < 0$ for $x > 1$.

$\therefore \lim_{x \rightarrow 1^+} \frac{1}{1-x} = -\infty$.

**Example 4.3**

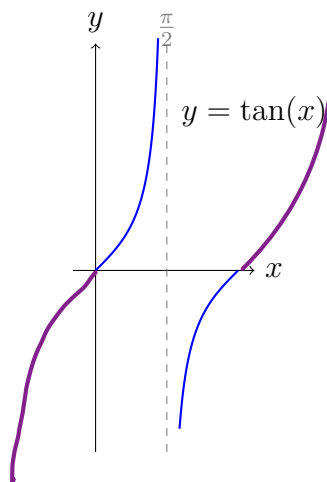
Find $\lim_{x \rightarrow 2^-} \frac{x^2 - 4x + 3}{(x-2)^3} = \infty$

Sol: $\lim_{x \rightarrow 2^-} x^2 - 4x + 3 = 4 - 8 + 3 = -1 < 0$

$\therefore \lim_{x \rightarrow 2^-} (x-2)^3 = 0$, $(x-2)^3 < 0$ if $x < 2$.

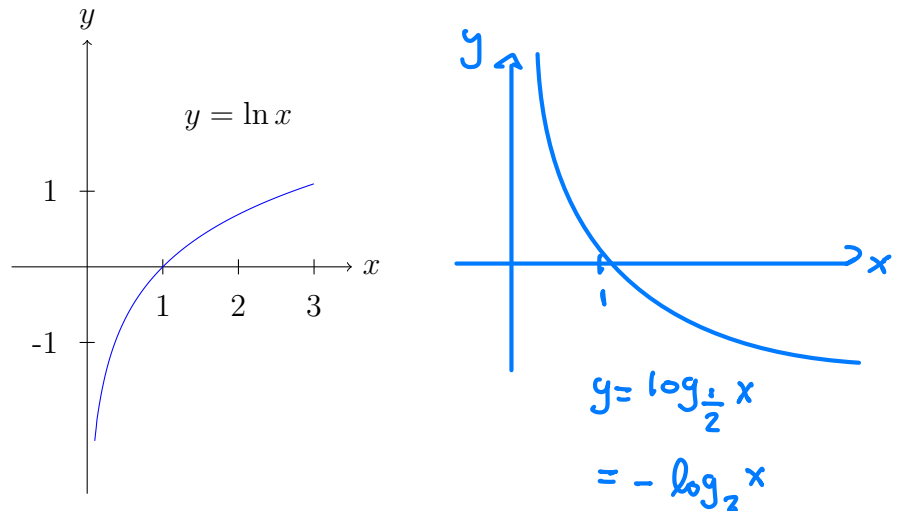
Example 4.4

Find $\lim_{x \rightarrow \frac{\pi}{2}^-} \tan x$ and $\lim_{x \rightarrow \frac{\pi}{2}^+} \tan x = -\infty$.

**Example 4.5**

Find $\lim_{x \rightarrow 0^+} \ln x = \lim_{x \rightarrow 0^+} \log_e x = -\infty$

$\lim_{x \rightarrow 0^+} \log_{\frac{1}{2}} x = \infty$

**Definition:**

The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statement is true:

$$\begin{aligned} \lim_{x \rightarrow a^-} f(x) = \infty & \quad \lim_{x \rightarrow a^-} f(x) = \infty & \quad \lim_{x \rightarrow a^+} f(x) = \infty \\ \lim_{x \rightarrow a^-} f(x) = -\infty & \quad \lim_{x \rightarrow a^-} f(x) = -\infty & \quad \lim_{x \rightarrow a^+} f(x) = -\infty \end{aligned}$$

Example 4.6

Find vertical asymptote(s) of $y = f(x) = \frac{x^2 - 4x + 3}{x^2 - x}$

Sol: Solve $x^2 - x = 0 \Rightarrow x = 0$ or 1 .

$$\lim_{x \rightarrow 1} \frac{x^2 - 4x + 3}{x^2 - x} = \lim_{x \rightarrow 1} \frac{(x-1)(x-3)}{x(x-1)} = \lim_{x \rightarrow 1} \frac{x-3}{x} = -2.$$

$\Rightarrow x=1$ is NOT a vertical asymptote.

$$\lim_{x \rightarrow 0^+} \frac{x^2 - 4x + 3}{x(x-1)} = -\infty \Rightarrow x=0 \text{ is a vertical asymptote.}$$

$$\left(\begin{array}{l} \because \lim_{x \rightarrow 0^+} x^2 - 4x + 3 = 3 > 0, \quad \lim_{x \rightarrow 0^+} x(x-1) = 0 \text{ and } x(x-1) < 0 \\ \text{when } 0 < x < 1 \\ \therefore \lim_{x \rightarrow 0^+} \frac{x^2 - 4x + 3}{x(x-1)} = -\infty. \end{array} \right)$$